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RFID tags

Abstract

A radio-frequency identification (RFID) tag for semi-permanent insertion into a gap in a storage grid includes an RFID antenna; a storage medium configured to store identification data, the storage medium being connected to the RFID antenna; and a case configured to house the RFID antenna and the storage medium. The case includes: an internal surface configured to support one or both of the RFID antenna and the storage medium; one or more feet configured to limit insertion of the case into the gap in the storage grid; and one or more protrusions configured to provide resistance to case movement relative the gap.

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Background/Summary

(1) This application claims priority from UK patent application number GB1900653.5, filed 17 Jan. 2019, the contents of which are incorporated herein by reference.
(2) The invention relates to radio-frequency identification (RFID) tags. In particular, it relates to RFID tags suitable for use with a storage and retrieval system.
BACKGROUND

(3) RFID tags may be used to store and transmit data, such as information relating to a product or device to which the RFID tag is attached or in which the RFID tag is embedded. An RFID tag may for example be configured to store identification data in the form of an identification number, a name or other information which enables the tagged product or device to be distinguished from other products or devices. The data may be transmitted to an RFID tag reader or scanner which receives the data, interprets the data and/or transfers the data to a further device for processing.
(4) The claimed RFID tags, methods, computer programs and systems are intended to provide improvements relative to known tags, methods, computer programs and systems.
SUMMARY
(5) According to an embodiment, there is provided a radio-frequency identification (RFID) tag as claimed in claim **1**. According to a further embodiment, there is provided a method as claimed in claim **8**. According to another embodiment, there is provided a computer program. According to a yet further embodiment, there is provided a system as claimed in claim **10**. Optional features are set out in the dependent claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The claimed RFID tags, methods, computer programs and systems will now be described in detail with reference to examples, in which:
(2) FIG. **1** schematically illustrates a storage grid of a storage and retrieval system;
(3) FIG. **2** schematically illustrates tracks of the storage grid of FIG. **1**;
(4) FIG. **3***a* schematically illustrates a robotic load-handling device for use with the storage grid of FIG. **1**;
(5) FIG. **3***b* schematically illustrates the robotic load-handling device of FIG. **3***a* in a cutaway view, with a container in a container-receiving space of the robotic load-handling device;
(6) FIG. **3***c* schematically illustrates the robotic load-handling device of FIG. **3***a* in a cutaway view, with a container being lowered from a container-receiving space of the robotic load-handling device;
(7) FIG. **4** schematically illustrates a storage grid of a storage and retrieval system, with a plurality of robotic load-handling devices as illustrated in FIGS. **3***a* to **3***c* on tracks of the storage grid;
(8) FIG. **5** schematically illustrates a first case member of an RFID tag in a top-perspective view;
(9) FIG. **6** schematically illustrates an RFID tag in a top-perspective view; and
(10) FIG. **7** schematically illustrates an RFID tag in a bottom-perspective view.
DETAILED DESCRIPTION
(11) The present embodiments represent the applicant's preferred examples of how to implement RFID tags, but they are not necessarily the only examples of how that could be achieved.
(12) A storage grid **1** of a storage and retrieval system is illustrated in FIG. **1**. The illustrated storage grid **1** includes a frame structure **14** comprising a plurality of upright members **16** that support horizontal members **18**, **20**. A first set of parallel horizontal members **18** is arranged orthogonally to a second set of parallel horizontal members **20** to form a plurality of horizontal grid structures supported by the upright members **16**. The members **16**, **18**, **20** are typically manufactured from metal. Containers **10** are stacked in substantially vertical stacks or columns **12** between the members **16**, **18**, **20** of the frame structure **14**, so that the frame structure **14** guards against horizontal movement of the stacks **12** of containers **10**, and guides or constrains vertical movement of the containers **10**.
(13) The illustrated storage grid **1** also includes a plurality of rails or tracks **22** arranged in a grid pattern above the stacks **12** of containers **10**, the grid pattern comprising a plurality of grid spaces, each stack **12** of containers **10** being located within a footprint of only a single grid space.

(14) FIG. **2** provides a plan view of the arrangement of the members **16**, **18**, **20** on the top layer of the storage grid **1**, showing the relative positions of the members **16**, **18**, **20** in more detail. In the illustrated embodiment, upright members **16** are arranged at four corners of a rectangle formed by parallel, neighbouring pairs of horizontal members **18** and parallel, neighbouring pairs of horizontal members **20**. The corresponding stack of containers **10** for each grid space is located within the rectangle defined by the horizontal members **18**, **20**. The illustrated pattern is repeated across some or all of the storage grid **1**. In the illustrated embodiment, the top-most horizontal members **18** of the storage grid **1** provide a first set of tracks **22***a* extending in a first direction (substantially along or parallel to the x-axis illustrated in FIG. **1** and FIG. **2**), and the top-most horizontal members **20** provide a second set of tracks **22***b* extending in a second, substantially orthogonal direction (substantially along or parallel to the y-axis illustrated in FIG. **1** and FIG. **2**). In some examples separate components may be placed on top of the top-most horizontal members **18**, **20** to provide the tracks **22***a*, **22***b*.
(15) Robotic load-handling devices (“robots” or “bots”) **30** having first and second sets of wheels **34** and **36** are configured to move laterally on the rails or tracks **22** above the stacks **12**, and to move containers **10** relative to the storage grid **1** using the bots' wheels **34** and **36** and respective container-lifting mechanisms which allow at least one container **10** to be lifted into a container-receiving space **40** within a body **32** of a respective bot **30**. The container-lifting mechanism may for example comprise one or more extendible and retractable vertical supports **38** which can be extended away from or retracted into the body **32** of the bot **30** to lower or raise a gripping device **39**. The gripping device **39** may be configured to engage the at least one container **10**, such that when the gripping device **39** is lowered or raised by the vertical supports **38** a gripped container **10** is correspondingly lowered or raised. An example of such a bot **30** is illustrated in FIG. **3***a*, with the bot's container-lifting mechanism shown in an extended configuration and engaging a container **10**. FIGS. **3***b* and **3***c* show the same bot **30** with part of its body **32** cut away to reveal the container-receiving space **40**, with (in FIG. **3***b*) the corresponding container-lifting mechanism in a retracted configuration (such that the corresponding container **10** is in a raised position, in the container-receiving space **40** in the body **32** of the bot **30**) and (in FIG. **3***c*) the corresponding container-lifting mechanism in an extended configuration (such that the corresponding container **10** is lowered, relative to the body **32** of the bot **30**).
(16) As illustrated in FIG. **4**, multiple bots **30** may be configured to move across the top of the frame structure **14** of a storage grid **1**, along tracks **22***a* and **22***b*. In order to coordinate the movement of multiple bots **30**, the bots **30** themselves and/or a control system configured to control one or more of the multiple bots **30** must know the bots' respective locations on top of the storage grid **1**. This location or position information may enable bots **30** to: travel to specific stacks **12** and retrieve particular containers **10**; to avoid collisions with other bots; and/or to avoid moving beyond the storage grid **1** and falling off the grid **1** (e.g. if, due to space, cost or other constraints, no barriers or other similar movement-constraining means are provided at the outer perimeter of the grid **1**).
(17) To provide a mechanism by which the bots **30** are able to determine where they are on the storage grid **1**, each bot may be provided with one or more RFID tag readers or scanners, and a plurality of RFID tags may be provided across the top of the storage grid **1**. As will be described in more detail below, the RFID tags may be inserted into suitably sized gaps in the storage grid **1**, such as gaps **24** provided in tracks **22***a*, **22***b* (see FIG. **2**).
(18) As the bots **30** move across the storage grid **1**, the bots' respective RFID tag readers may read signals from one or more of the RFID tags as the bots pass the RFID tags.
(19) The RFID tags may be fixed relative to the storage grid **1**, such that the RFID tags' positions do not change, and thus provide a series of fixed reference points for the bots **30**. The positions of the RFID tags may for example be recorded in a map, database or other record which can be consulted by a processor. The positions of the RFID tags may be defined in various different ways,

such as in terms of distances along the tracks **22***a*, **22***b*, and/or in terms of which grid space or grid spaces they coincide with or are closest to. For example, if an RFID tag is located in a gap **24** between tracks **22***a* or **22***b*, the record may include identification data for one or both of the two grid spaces that the RFID tag is located between. As another example, if the RFID tag is located at an outer edge of a grid space at the outer edge of the storage grid **1**, the record may include identification data for only the single grid space which the RFID tag is located at or adjacent to. As a further example, if the RFID tag is located at or close to a junction or crossroads between tracks **22***a* and **22***b*, the record may include identification data for one or more of the two or four grid spaces that meet at the junction, and/or identification data for the corresponding upright member **16**. As a still further example, a location of an RFID tag may be identified in the record with one or more measurements (e.g. in metres, yards or feet) taken from a known “origin” of the storage grid **1** (e.g. a known corner, an edge or a central point).

(20) As explained, there may be various ways that a particular location of an RFID tag on a storage grid **1** may be recorded. In some examples, the location data (identifying the location of the RFID tag) may be encoded in a storage medium of the RFID tag and be transmittable to an RFID tag reader which comes sufficiently close to the RFID tag.

(21) In other examples, the RFID tag may contain data which enables a particular RFID tag to be identified (e.g. a character string which is particular to that RFID tag), and the location data may be stored in a record against the RFID tag's identification data. The process of installing RFID tags on a storage grid **1** may include a process of writing the location data to the RFID tags after installation (e.g. when their positions are known). In other examples, the process of installing the RFID tags may require the installer to install specific RFID tags at specific, pre-determined locations, and/or to record the locations that specific RFID tags are installed at. In some examples, a storage medium of an RFID tag may be encoded with both location data and data which enables the particular RFID tag to be identified.

(22) The bots **30** may use information read from one or more RFID tags they have encountered while moving across the grid **1** to determine where they are on the grid **1**.

(23) The number of RFID tags provided across the storage grid **1** may determine the precision with which the bots **30** can locate themselves, and/or the speed with which they can determine their locations and/or their directions of travel. For example, if at least one RFID tag is provided for each grid space in the storage grid **1**, a bot **30** may be able to determine its position relatively quickly and precisely, whereas if RFID tags are provided only for every second or third grid space, the bot **30** may need to move several grid spaces before being able to identify where it is going. In some examples, an RFID tag may be provided along each side of each four-sided grid space (possibly excluding edge cells, which may have fewer RFID tags). In other examples, RFID tags may be provided only along one, two or three sides of each grid space, and/or only at corners or junctions between grid spaces. One or more RFID tag readers may be mounted on each bot **30** in such a position that the RFID tags on the storage grid **1** can be read by the RFID reader(s) as the corresponding bot **30** travels along the corresponding sections of tracks **22***a*, **22***b*. For example, if the bots' RFID tag readers are provided at bottom-right corners of the bots **30**, RFID tags may correspondingly be provided at bottom-right corners of grid spaces. The numbers of RFID tags per grid space and the number of RFID tag readers per bot **30** may be chosen to balance the cost of additional tags and/or readers against the reliability of the individual tags and/or readers and/or the need for redundancy and/or against the need for quickly identifying which direction a bot **30** is moving in.

(24) FIGS. **5** to **7** illustrate an example of an RFID tag **61**. FIG. **5** shows a first case member **41** of the RFID tag **61**. The first case member **41** includes a surface **43** which is arranged to support one or both of an RFID antenna and a storage medium of the RFID tag **61**. The RFID antenna and the storage medium may for example be mounted on one or more printed circuit boards (PCBs), one or more plastic inlays and/or one or more other supports which may be mounted on the surface **43**.

The RFID antenna may, when an RFID tag reader passes within sufficiently close proximity and/or remains within sufficiently close proximity for a sufficient period of time, transmit identification data stored on the storage medium to the nearby RFID tag reader. The RFID antenna and storage medium may preferably be a passive arrangement, i.e. one requiring no direct power source and instead relying on electromagnetic waves received at the antenna (e.g. from a nearby RFID tag reader) to cause and/or facilitate transmission of the identification data. In other examples, the RFID antenna and storage medium may be an active arrangement, i.e. one with a direct power source of its own, the direct power source powering transmission of the identification data. A passive arrangement may advantageously provide greater freedom of location of the RFID tag **61** on the storage grid **1** and/or reduced size of the RFID tag **61**, since no power cabling or battery needs to be provided for the RFID tag **61**. A passive arrangement may also advantageously reduce the cost of manufacture and/or ongoing use of the RFID tag **61**, and/or simplify the installation of the RFID tag **61**, since the RFID tag **61** need only be positioned at the desired point on the storage grid **1**—no further equipment or installation is involved. An active arrangement may advantageously provide greater flexibility regarding the position of the RFID tag **61** within each grid space and/or the position of the RFID tag reader on the bot **30**, since the antenna of the RFID tag **61** may be capable of transmitting the identification data further than an antenna of a passive arrangement would be able to.

(25) The first case member **41** also includes a plurality of feet **45** which are configured to limit insertion of the RFID tag **61** into a gap **24** in the storage grid **1**. The feet **45** may for example protrude in the negative z-direction to a specific extent to provide a known minimum distance between the effective contact surface of the bottom of the first case member **41** (i.e. the bottoms of the feet **45**) and the internal surface **43** which is configured to support the RFID antenna. This may advantageously improve the probability that the RFID antenna will be and remain sufficiently close to an RFID tag reader of a passing bot **30** that the RFID tag reader can read data transmitted by the RFID antenna of the RFID tag **61**. This may be particularly advantageous in environments where the gap **24** in the storage grid **1** comprises a metal material which may interfere with or attenuate RFID signals transmitted by the RFID antenna, such that it would be disadvantageous for the RFID tag **61** to be forced too far into the gap **24**.

(26) The feet **45** may also provide a known distance in the positive z-direction between the effective contact surface of the bottom of the first case member **41** (the bottoms of the feet **45**) and an upper surface or greatest z-height of the RFID tag **61**. This may significantly improve the ease and speed of installation of RFID tags in a storage grid **1**, as well as the reliability of the RFID tag **61** being able to communicate data to an RFID tag reader of a bot **30**, since an RFID tag installer can push the RFID tag **61** into a gap **24** until the feet **45** contact a bottom surface of the gap **24**—the installer does not need to perform any measurements or adjustments to ensure that the RFID tag **61** will be within reading distance of a passing RFID tag reader or to ensure that the RFID tag **61** does not protrude sufficiently far that it will be struck and potentially damaged or displaced by the wheels **34**, **36** of a bot **30** passing on the tracks **22***a* or **22***b* between which the RFID tag **61** is positioned (in gap **24**).

(27) The feet **45** may furthermore provide a protective and/or reinforcing function to help minimise the risk of the RFID tag **61** and its internal components (e.g. RFID antenna and storage medium) being crushed by the wheels **34**, **36** of a passing bot **30**. The feet **45** may for example effectively extend in the z-direction from the bottom to the top of the first case member **41**, to provide relatively robust and/or rigid supports for the first case member **41**. The feet **45** (and possibly also the first case member **41**) may preferably be made of a relatively rigid material, such as a polycarbonate material, to provide a relatively inflexible structure. The feet may be integrally formed with the first case member **41** or may be attached to it. The feet may in some embodiments extend upwards in the z-direction beyond the internal surface **43** to provide additional protection for the internal surface **43** and any components mounted on the internal surface **43**, e.g. by

providing one or more features or surfaces which a wheel **34**, **36** of a bot **30** may contact before and in preference to one or more components mounted on the internal surface **43**.
(28) FIG. **6** illustrates a second case member **63** which has been overmoulded onto the first case member **41** illustrated in FIG. **5** to create a case which substantially surrounds the RFID antenna and the storage medium supported by the surface **43** of the first case member **41**. The second case member **63** includes a plurality of protrusions **65** which protrude outwards from the two longer sides of the second case member **63** illustrated in FIG. **6**. The illustrated protrusions **65** extend along most of the vertical height of the two longer side walls and are tapered at a lower end. The taper at the lower end may help facilitate insertion of the RFID tag **61** into a gap **24**. The protrusions **65** are configured to provide resistance to movement of the second case member **63** (and thus also the first case member **41** onto which the second case member **63** is overmoulded) within a gap **24**. The protrusions may comprise a relatively deformable material which can be deformed by application of force to insert the RFID tag **61** (including the first and second case members **41**, **63**) into a gap **24**. The protrusions **65**, once deformed by contact with walls of a gap **24** as described above, may then provide resistance to movement of the RFID tag **61** along the gap **24** (i.e. in the x- or y-directions illustrated in FIG. **1** and FIG. **2**) and/or into or out of the gap **24** (i.e. in the z-direction illustrated in FIG. **1** and FIG. **2**). The deformation of the protrusions **65** may lead to an increased surface area of the second case member **63** being in contact with walls of a gap **24**, and therefore to increased friction between the second case member **63** and the walls of the gap **24**. This may advantageously help to ensure that the RFID tag **61** is not displaced from its position of installation, e.g. by a passing bot **30** if the bot **30** comes into direct contact with the RFID tag **61** or by vibration of the grid **1** due to movement of one or more bots **30** along the tracks **22***a*, **22***b* provided on the storage grid **1**. The second case member **63** and the protrusions may preferably comprise a thermoplastic material to provide a relatively deformable structure over the relatively rigid and inflexible structure of the first case member **41**.
(29) Preferably materials are chosen for the first case member **41** and the second case member **63** which do not significantly expand or contract with temperature changes.
(30) In some embodiments, one or more protrusions may also or alternatively be provided on the other (shorter) sides of the second case member **63**, e.g. if the RFID tag **61** is to be inserted into a space which is enclosed on more sides than the illustrated gaps **24** are.
(31) The first case member **41** illustrated in FIG. **5** additionally includes features **47**, **49**. The features **47**, **49** may serve one or more purposes. For example, one or more of the features **47**, **49** may be alignment features configured to facilitate alignment of overmoulding apparatus with the first case member **41**. For example, the illustrated features **47** may be configured to allow an overmoulding apparatus to hold or support the first case member **41** using the features **47**, such that the overmoulding apparatus can overmould the second case member **63** over the first case member **41**. The features **47** may for example be sized and/or positioned such that at least part of one or more of the features **47** will not be directly overmoulded, such that the overmoulding apparatus can continue to contact and support the first case member **41** while the second case member **63** is overmoulded over the first case member **41**. Similarly, the illustrated features **49** may be provided to help with alignment of the first case member **41** and an overmoulding apparatus and/or to help ensure that, after the overmoulding process has been completed, the first and second case members **41**, **63** cannot move relative to one another. For example, the features **49** may be configured to protrude into or through corresponding apertures which are to be formed in the second case member **63**, thus constraining relative movement of the first and second case members **41**, **63**. The first and second case members **41**, **63** together form a case of the illustrated RFID tag **61**, the case defining an interior within which the internal surface **43** is located, the RFID antenna and the storage medium being supported by the internal surface **43**.
(32) FIG. **7** shows, in perspective view, the underside of the RFID tag **61** after the overmoulding process has been completed. In the illustrated example, the bottoms of the feet **45** of the first case

member **41** are flush with the bottom of the overmoulded second case member **63** (i.e. flush in the z-direction). In other examples, the feet **45** may protrude beyond the bottom of the overmoulded second case member **63** (i.e. beyond the bottom surface of the second case member **63**, in the negative z-direction), or be slightly recessed inside the overmoulded second case member **63** (i.e. above the bottom of the overmoulded second case member **63**, in the positive z-direction). In all cases, the feet **45** serve to limit the extent to which the RFID tag **61** can be inserted into a gap **24**, by providing a relatively rigid, incompressible structure which can limit or prevent further insertion into a gap **24** by relatively incompressibly abutting a bottom surface of the gap **24**. If the feet **45** are recessed inside the overmoulded second case member **63** and the overmoulded second case member **63** comprises a resiliently deformable material which deforms as an installation force is applied to the RFID tag **61**, the feet **45** will still ensure that the minimum distance between the surface **43** and the bottom contact surface of the RFID tag **61** is maintained.
(33) FIG. **7** shows a further set of features **51** in the form of holes **51**. Like features **47**, **49**, the holes **51** may be used to facilitate the overmoulding of the second case member **63** over the first case member **41**. For example, the holes **51** may extend through the bottom of the second case member **63** and the first case member **41**, allowing supports to be inserted into the first and second case members **41**, **63** and to support both of the case members **41**, **63** as the overmoulding process is completed.
(34) Advantageously, an RFID tag **61** with a case comprising relatively rigid, inflexible material configured to support the RFID antenna and storage medium and to resist forcing of the RFID tag **61** beyond a bottom surface of a gap **24** and a relatively deformable material including protrusions configured to deform to resist movement of the RFID tag **61** relative to the gap **24** in the storage grid **1** may provide a robust and easy-to-install RFID tag **61** which is capable of withstanding crushing forces (e.g. if run over by the wheels **34**, **36** of a bot **30**) due to the deformability of the relatively deformable material (i.e. the overmoulded second case member **63** in the illustrated examples) over the relatively rigid material (i.e. the first case member **41** in the illustrated examples) and which is capable of resisting forces which would move the RFID tag **61** within or out of a gap **24** after insertion (e.g. vibration or crushing forces due to the movement of bots **30** over the storage grid **1**), thereby providing an RFID tag **61** which continues to reliably perform its intended function of providing a fixed reference point for moving bots **30**.
(35) Advantageously, the illustrated RFID tag **61** may be relatively inexpensive to manufacture, since the overmoulding of the second case member **63** over the first case member **41** may be automatable, and thus the process of securing the RFID antenna and storage medium within the overall case formed by the first and second case members **41**, **63** may require minimal human input. The illustrated RFID tag **61** may furthermore help ensure that an RFID signal can always be received from the RFID tag **61** by helping to maintain the RFID antenna and storage medium a consistent height above the bottom of the gap **24** in the storage grid **1** and therefore at a consistent distance from bots **30** moving over the top of the storage grid **1**. The illustrated RFID tag **61** moreover advantageously does not require an adhesive to affix the RFID tag **61** in place relative to the storage grid **1**, since the protrusions **65** formed of a deformable material deform to increase the surface area of the RFID tag **61** that is in contact with the walls defining the gap **24**, thereby increasing the friction between the RFID tag **61** and the gap **24** and holding the RFID tag **61** in place. The RFID tag **61** also does not require an installation tool to install it in a gap **24** in a storage grid **1**—the RFID tag **61** can be pushed into the gap **24** by hand.
(36) As previously discussed, multiple RFID tags **61** may be distributed across a frame structure **14** of a storage grid **1** to provide, in conjunction with RFID tag readers provided on one or more bots **30**, a system which allows the bots **30** and/or a controller of one or more of the bots **30** to determine the bots **30** locations on the frame structure **14** of the storage grid **1**.
(37) In some embodiments, the illustrated and described system of RFID tags **61** on a storage grid **1** and RFID tag readers mounted on bots **30** moving across a storage grid **1** may be one of several

systems or data sources which bots **30** and/or a controller of bots **30** can use to determine locations of bots **30** on a storage grid **1**. For example, a further system may be provided which relies on a different mechanism (e.g. a GPS-like or mesh network-like system which determines positions using triangulation of signals received from different transmitters or beacons) to determine the location of one or more bots **30** on a storage grid **1**. In such cases, the system of RFID tags **61** and RFID tag readers may be used to provide an initial or confirmatory indication of a bot's location, and/or to provide a location if the other system is unable to do so, and/or to provide a faster, more reliable or more precise indication of a bot's location.

(38) The illustrated and described RFID tags **61** may therefore, by virtue of their features (including feet **45** and protrusions **65**), be suitable for semi-permanent insertion into a gap **24** in a storage grid **1** to allow the movement of bots **30** to be monitored using the RFID tags **61** over an indefinite period of time. The RFID tags **61** may, due to their robustness of construction (including relative rigidity of feet **45** and relative deformability of protrusions **65**) be capable of relatively long service in the environment of a storage grid **1** where the RFID tags **61** may be vibrated or struck by moving bots **30**. The RFID tags **61** may for example be capable of operating (and remaining substantially spatially fixed) in the environment of a storage grid **1** for a number of weeks, months or even years, such as over a significant portion of the expected lifetime of a storage grid **1**. This may advantageously minimise time spent reinstalling RFID tags on a storage grid **1**, which may help to maximise "uptime" of the storage grid **1**, i.e. time during which the storage grid **1** is in use, with one or more bots **30** operating on top of the storage grid **1** to move containers between locations.

(39) Although in the illustrated embodiment the case is provided by two case members **41**, **63**, in other embodiments, the case may be a single-member case. In further embodiments there may be more than two case members which collectively form the case. Furthermore, two or more case members may be connected or attached to one another in one or more different ways—they need not necessarily be attached by an overmoulding process.

(40) Although in the illustrated embodiment the protrusions **65** are provided on the second case member **63**, in other embodiments one or more of the protrusions **65** may be provided on the first case member **41** (i.e. on the same case member as the feet **45**).

(41) In such embodiments, the second case member **63** may have the primary purpose of covering or enclosing the internal surface which is configured to support the RFID antenna and/or the storage medium. Conversely, in some embodiments both the feet **45** and the protrusions **65** may be provided on the second case member **63**. In other embodiments, the case may comprise only a single member, in which case both the feet **45** and the protrusions **65** may be provided on the same, single case member, with no further case members for enclosing the RFID antenna and/or storage medium—the single case member may itself provide the internal surface **43** without requiring enclosing by another case member. In further embodiments, the case may comprise more than two members, in which case the feet **45** and/or the protrusions **65** may be provided on one or more of the different case members.

(42) Although in the illustrated embodiments the feet **45** have rounded rectangular or rectelliptical cross sections, in other embodiments the feet may take different shapes or structures. For example, in some embodiments, the feet may have rectangular, circular, elliptical, triangular, square or any other cross sections. Additionally, the cross sections of different feet may be different from each other. Furthermore, although in the illustrated embodiments the feet **45** are solid in cross section, in other embodiments one or more of the feet may be at least partially hollow in cross section, e.g. to provide within a given foot a hole such as the holes **51** illustrated in FIG. **7**. In some embodiments, the feet may not have constant cross sections along their entire z-lengths. For example, in some embodiments, the feet may be deliberately tapered, or splayed at the lower (in the z-direction) ends, e.g. to provide a specific surface area for contacting the bottom of a gap **24**.

(43) Although in the illustrated embodiments there are three feet **45**—one at or near each end

(along the x-axis) of the RFID tag **61**, and one in or near the middle (along the x-axis)—in other embodiments more or fewer feet may be provided. For instance, it may be advantageous for more or fewer feet to be provided based on the cross-sectional shape and/or other features of the feet. In some embodiments, for example, the feet may take the form of rods (e.g. cylindrical, triangular or square rods) with relatively narrow cross sections, in which case it may be preferable for more feet to be provided. The position of the feet along the x- and y-axes of the first case member **41** and/or the lengths of the feet in the z-axis may be determined based upon a desired effect of the feet. For example, it may be preferable for the feet to be relatively evenly spaced along the x- and y-axes and for the feet to be substantially identical in length along the z-axis if the desired effect is for the RFID tag **61** to sit squarely within a gap **24**, with the bottoms of the feet at substantially the same z-axis height as each other and the top of the RFID tag **61** at substantially the same z-axis height along the x- and y-axes of the RFID tag **61**. In other examples, it may be preferable for the feet to have different lengths, e.g. if the desired effect is for a first x-direction end of the RFID tag **61** to have a relatively large z-axis height relative to the opposite x-direction end of the RFID tag **61**, e.g. if the RFID antenna is mounted at the first end of the RFID tag **61** and it is desired to increase the probability of the RFID antenna being within transmitting/receiving distance of a passing bot's RFID tag reader.

(44) Although in the illustrated embodiments there are five protrusions **65** on each of the longer sides of the RFID tag **61**, in other embodiments there may more or fewer protrusions. As mentioned above, in some embodiments one or more protrusions may additionally be provided on the shorter sides of the RFID tag **61**. The number of protrusions and/or the x- and/or y-widths of the protrusions may be chosen to try to optimise resistance to movement of the RFID tag **61** within a gap **24** while also allowing initial insertability of the RFID tag **61** into the gap **24**. Although the illustrated protrusions extend substantially in the z-direction (i.e. vertically along the longer sides of the RFID tag **61**), in other embodiments one or more of the protrusions may extend at least partially in a different direction. For example, it may be advantageous in some embodiments for one or more of the protrusions to extend at an angle to the vertical (i.e. at an angle to the z-axis), and/or to change direction of extension along the length of the protrusion, e.g. by being angled or curved.

(45) Although the illustrated protrusions include a substantially-constant-y-width section above the tapered section, in other embodiments one or more of the protrusions may have variable widths, e.g. to create a ribbed or otherwise variable outer profile of the protrusion(s). This may for example help to increase the ability of the protrusion(s) to resist movement of the RFID tag **61** within a gap **24** by increasing traction between the walls of the gap **24** and the protrusions.

(46) In some embodiments, the protrusions may be configured to engage with specific features on walls of a gap **24**, e.g. corresponding grooves or recesses. This may help to optimise resistance to movement of the RFID tag **61** within the gap **24**.

(47) It is envisaged that any one or more of the variations described in the foregoing paragraphs may be implemented in the same embodiment of an RFID tag.

(48) In this document, the language "movement relative to a gap" is intended to include movement within the gap, e.g. sliding along the gap, as well as movement into or out of a gap.

(49) In this document, the language "movement in the n-direction" (and related wording), where n is one of x, y and z, is intended to mean movement substantially along or parallel to the n-axis, in either direction (i.e. towards the positive end of the n-axis or towards the negative end of the n-axis).

(50) In this document, the word "connect" and its derivatives are intended to include the possibilities of direct and indirection connection. For example, "x is connected to y" is intended to include the possibility that x is directly connected to y, with no intervening components, and the possibility that x is indirectly connected to y, with one or more intervening components. Where a direct connection is intended, the words "directly connected", "direct connection" or similar will be

used. Similarly, the word “support” and its derivatives are intended to include the possibilities of direct and indirect contact. For example, “x supports y” is intended to include the possibility that x directly supports and directly contacts y, with no intervening components, and the possibility that x indirectly supports y, with one or more intervening components contacting x and/or y.
(51) In this document, the word “comprise” and its derivatives are intended to have an inclusive rather than an exclusive meaning. For example, “x comprises y” is intended to include the possibilities that x includes one and only one y, multiple y's, or one or more y's and one or more other elements. Where an exclusive meaning is intended, the language “x is composed of y” will be used, meaning that x includes only y and nothing else.

Claims

1. A radio-frequency identification (RFID) tag for semi-permanent insertion into a gap between first and second tracks in a storage grid, the RFID tag comprising: an RFID antenna; a storage medium configured to store identification data, the storage medium being connected to the RFID antenna; and an elongated case configured to house the RFID antenna and the storage medium, the elongated case including: a first case member including an internal surface configured to support one or both of the RFID antenna and the storage medium, and one or more feet configured to limit insertion of the elongated case when inserted into the gap between the first and second tracks in the storage grid; and a second case member including one or more protrusions configured to provide resistance to movement of the elongated case relative to the gap between the first and second tracks in the storage grid when inserted into the gap.
2. An RFID tag as claimed in claim 1, wherein the one or more protrusions comprise: resiliently deformable protrusions configured to be deformed to allow the elongated case to be inserted into the gap between the first and second tracks in the storage grid, and configured to provide resistance to movement of the elongated case out of the gap between the first and second tracks when the case has been inserted into the gap.
3. An RFID tag as claimed in claim 1, wherein each of the one or more protrusions comprises: a tapered section configured to facilitate insertion of the elongated case when inserted into the gap between the first and second tracks in the storage grid.
4. An RFID tag as claimed in claim 1, wherein the one or more feet comprise: rigid feet configured to provide a minimum distance between an effective contact surface of the one or more feet and the internal surface.
5. An RFID tag as claimed in claim 1, comprising: a printed circuit board or a plastic inlay wherein the RFID antenna and the storage medium are mounted on the printed circuit board or the plastic inlay which is supported by the internal surface of the first case member.
6. An RFID tag as claimed in claim 1, wherein the first case member comprises a rigid material, and the second case member comprises a deformable material, wherein the second case member is overmoulded onto the first case member.
7. An RFID tag as claimed in claim 6, wherein the first case member includes one or more additional features configured to facilitate one or more of: supporting the first case member during an overmoulding process; alignment of an overmoulding apparatus with the first case member; and alignment of the second case member with the first case member.
8. A method of manufacturing a radio-frequency identification (RFID) tag for semi-permanent insertion into a gap between first and second tracks in a storage grid, the method comprising: providing an RFID antenna; providing a storage medium configured to store identification data, the storage medium being connected to the RFID antenna; and providing an elongated case configured to house the RFID antenna and the storage medium, the case includes: a first case member including an internal surface configured to support one or both of the RFID antenna and the storage medium and one or more feet configured to limit insertion of the case into the gap between the first

and second tracks in the storage grid; and a second case member including one or more protrusions configured to provide resistance to movement of the case relative to the gap between the first and second tracks in the storage grid.

9. A method as claimed in claim 8, further comprising: overmoulding the second case member over the first case member, and wherein providing the first case member comprises: providing one or more additional features on the first case member, the one or more additional features including at least an alignment feature configured to align the first case member with the second case member and align the first case member during an overmoulding process, and wherein overmoulding the second case member comprises: aligning an overmoulding apparatus with the first case member using the one or more additional features.

10. A system comprising: a storage grid having first tracks and second tracks; and a plurality of RFID tags, each RFID tag including: an RFID antenna; a storage medium configured to store identification data, the storage medium being connected to the RFID antenna; and an elongated case configured to house the RFID antenna and the storage medium, the elongated case including: a first case member including an internal surface configured to support one or both of the RFID antenna and the storage medium, and one or more feet configured to limit insertion of the case when inserted into a gap between the first tracks and the second tracks in the storage grid; and a second case member including one or more protrusions configured to provide resistance to movement of the case relative to the gap between the first tracks and the second tracks in the storage grid when inserted into the gap.

11. The system as claimed in claim 10, comprising: a plurality of robotic load-handling devices.

12. The system as claimed in claim 11, wherein each robotic load-handling device comprises: an RFID tag reader.